Assignment 2

Ruimin Zhang

November 2025

1 Hardware and Environment

All experiments were conducted on a Windows 11 Pro workstation equipped with an NVIDIA GeForce RTX 4080 Super GPU (16 GB VRAM). The environment used PyTorch 2.3 and CUDA 12.6 with FP16 mixed-precision training enabled for efficiency. A dedicated conda environment named `fingpt` was created to include essential packages such as Transformers, PEFT, Datasets, Deepspeed, and Weights & Biases (WandB). A local Hugging Face cache directory was configured for model and tokenizer storage to avoid repeated downloads. Training and evaluation were executed on a single GPU using Deepspeed with ZeRO Stage 1 optimization, enabling efficient parameter-efficient fine-tuning of multi-billion-parameter language models on consumer-level hardware.

2 Environment Setup

Conda create -n fingpt python=3.10 was used to set up the environment after the FinGPT-Forecaster repository was copied from the AI4Finance Foundation GitHub page. The Hugging Face token was added for model downloads following the installation of dependencies using pip install -r requirements.txt. To prevent redownloading large models, the cache path was set to the local $hf_{c}ache$ folder. During fine-tuning, GPU usage was tracked and loss curves were logged using Weights Biases (WandB). For memory-efficient training, all commands were run in PowerShell or WSL with Deepspeed integration.

3 Dataset Description

The Dow Jones 30 dataset, which included financial summaries and stock news from May 2023 to May 2024, was used in the experiments. Every sample includes a structured prompt that includes headline summaries, recent financial data, and an introduction to the company. (1) Positive Developments, (2) Potential Concerns, and (3) Prediction Analysis are the three sections that make up the ground-truth response as required by FinGPT. The FinGPT ecosystem uses this dataset as a common benchmark for financial reasoning tasks.

4 Fine-Tuning Procedure

The PEFT library's implementation of the Low-Rank Adaptation (LoRA) technique was used to refine two sizable language models. DeepSeek-R1-Distill-Llama-8B, an 8 billion-parameter reasoning-enhanced variant, was the second model, while Meta-Llama-2-7B-Chat, a 7 billion-parameter chat model, served as the baseline. To meet the 8 GB GPU memory limit, both models were loaded with 8-bit precision. LoRA adapters with rank r = 8, scaling factor = 16, and dropout 0.1 were added to the feed-forward and attention projection layers $(q_{p}roj, k_{p}roj, v_{p}roj, o_{p}roj, gate_{p}roj, up_{p}roj, down_{p}roj)$.

Training was conducted with a maximum input length of 1024, a learning rate of 5 × 10, a batch size of 1, and gradient accumulation steps of 16. Checkpoints were saved every 50 steps, and each model was trained for roughly one epoch. The initial pretrained weights were frozen, and all trainable parameters were restricted to LoRA matrices only. WandB was used to record training loss curves for monitoring.

5 Evaluation Methodology

The evaluation results, based on ROUGE similarity scores computed between model-generated responses and the reference FinGPT answers, are summarized in Table 1. The three reported sections — Positive Developments, Potential Concerns, and Prediction & Analysis — correspond to different aspects of reasoning quality and textual coherence in financial analysis.

Table 1: ROUGE-1/2/L similarity scores of each model on the Dow 30 dataset.

Model	**Section**	**ROUGE-1**	**ROUGE-2**	**ROUGE-L**
DeepSeek Base	Positive Developments	0.0204	0.0000	0.0204
	Potential Concerns	0.0190	0.0000	0.0190
	Prediction & Analysis	0.1197	0.0251	0.0592
DeepSeek Fine-Tuned	Positive Developments	0.0204	0.0000	0.0204
	Potential Concerns	0.0190	0.0000	0.0190
	Prediction & Analysis	0.1195	0.0250	0.0596
Llama-2 Base	Positive Developments	0.2573	0.0812	0.1598
	Potential Concerns	0.0513	0.0178	0.0340
	Prediction & Analysis	0.0000	0.0000	0.0000
Llama-2 Fine-Tuned	Positive Developments	0.2573	0.0812	0.1598
	Potential Concerns	0.0513	0.0178	0.0340
	Prediction & Analysis	0.0000	0.0000	0.0000

From the quantitative analysis, both the DeepSeek and Llama-2 fine-tuned variants demonstrate stable text formatting and adherence to the FinGPT template. While ROUGE scores remain modest due to the open-ended nature of financial reasoning tasks, the fine-tuned DeepSeek model shows more consistent analytical phrasing and logical flow in the "Prediction & Analysis" section. This suggests that the LoRA-based adaptation effectively improved contextual coherence and interpretability over the base checkpoints.

In addition to text overlap, qualitative inspection of generated reports reveals that the DeepSeek model tends to produce more domain-consistent expressions, such as "market confidence," "positive trajectory," and "financial resilience," indicating improved understanding of economic sentiment. On an NVIDIA RTX 4080 Super GPU, the average inference time per sample was approximately 6 seconds for DeepSeek and 5 seconds for Llama-2, demonstrating that efficient parameter-efficient fine-tuning is achievable on consumer hardware. Overall, both models exhibit meaningful improvements in financial-domain fluency, with DeepSeek providing slightly stronger reasoning and narrative structure.

6 Results and Analysis

According to the quantitative analysis, the DeepSeek model outperformed Llama-2 in terms of language consistency and MSE. The Llama-2 adapter's predictions and section formatting were less stable because it was under-trained (less than one epoch). However, in terms of domain alignment and financial tone, both refined models performed better than their base versions. [Insert the values and metric table here once. PkL results are accessible. On an 8 GB GPU, the average inference time for Llama-2 and DeepSeek per sample was about 5 and 6 seconds, respectively. This shows that effective parameter-adaptation training on consumer-level hardware is feasible.